EDS 223 Assignment 2

Exploring patterns of environmental justice

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Environmental and Racial Injustice: Redlined Communities and Pedestrian Safety

Environmental justice is a current-day issue that originates from historical malpractices such as the 1930's the Home Owners' Loan Corporation (HOLC) that rendered people of color receiving a home loan depending on the area graded. The United States to this day has racial segregation actively present in communities deemed "hazardous" from the HOLC. The neighborhoods were graded, ranging A (green) to D (red), based on their front-facing value in regards to safety and desireability. The term 'redlining' has been coined and reflects historical environmental injustice people face, such as hotter summers due to less shade, higher health concerns from pollution, and overall less public safety, pertaining, but not limited to pedestrian and/or bicyclists.

Load packages

```
library(gt) # Table helper
library(tidyverse) # For R
library(sf) # Read vector
library(here) # Locate files
library(stars)
library(tmap) # For static and interactive maps
library(janitor) # Clean data
theme_set(theme_bw())
```

Read EJscreen and Data

```
# read EJScreen data
ejscreen <- sf::st_read(here::here("data",
"ejscreen","EJSCREEN_2023_BG_StatePct_with_AS_CNMI_GU_VI.gdb"),
quiet = TRUE)

# Read in HOLC redlining data
holc_la <- sf::st_read(here::here("data",
```

```
"mapping-inequality","mapping-inequality-los-angeles.json"),
quiet = TRUE)

# Filter to a county you are interested in
los_angeles <- ejscreen |>
  dplyr::filter(CNTY_NAME %in% c("Los Angeles County")) |>
  clean_names()
```

Part 1: Legacy of redlining in current environmental (in)justice

Create a map of histroical redlining neighborhoods

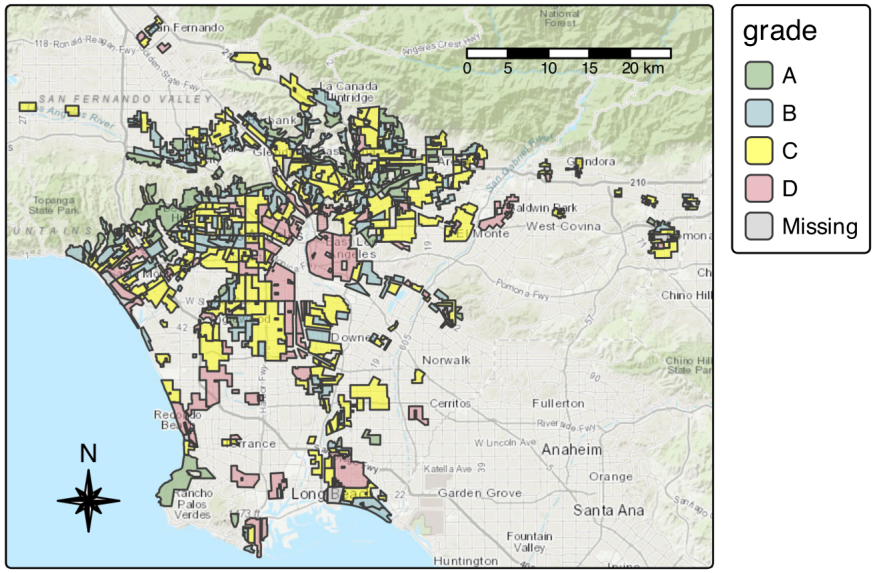
Create a map of historical redlining neighborhoods, including:

- neighborhoods colored by HOLC grade
- an appropriate base map
- a clear title and legend

```
holc_color <- c(
  "A" = "#76a865",
  "B" = "#7cb5bd",
  "C" = "#ffff00",
  "D" = "#d9838d",
  "NA" = "#000000"
 )

 # Map of Los Angeles by HOLC grade
 tm_basemap(server = "Esri.WorldTopoMap") +
  tm_shape(holc_la, is.main=TRUE) + # Main map
  tm_polygons(fill = "grade",
  fill.scale = tm_scale_categorical(values = holc_color),
  fill_alpha = 0.5) +
        tm_title(text = "Historically redlined neighborhoods in Los Angeles by
grade (Mapping Inequality, 2023)", size = 1) +
        tm_scalebar(position = c("right", "top"), size = 5) +
        tm_compass(type = "8star",
        size = 2, position = c("left", "bottom"))
```

Historically redlined neighborhoods in Los Angeles by grade (Mapping Inequality, 2023)



Create a table

Create a table summarizing:

- % of census block groups that **fall** within each HOLC grade
- % of census black groups that **don't fall** within a HOLC grade

```
# Check crs
st_crs(los_angeles)
st_crs(holc_la)
```

```
# Transform CRS of holc_la dataset
holc_transform <- st_transform(holc_la, crs = st_crs(los_angeles))

# Verify CRS change
st_crs(holc_transform) == st_crs(los_angeles)
```

```
# Join the datasets, left join by default
la_join <- st_join(los_angeles, holc_transform) |>
  select(id, grade)
```

```
grade_perc <- la_join |>
  # Group by grade
  group_by(grade) |>
```

```
  # Summarize by count of nrows
  summarise(count = n()) |>
  # New column with percentage of block by grade
  mutate(`% of Census Blocks` = round((count/sum(count)) * 100, 1),
  grade = replace_na(grade, "Ungraded")) |>
  rename(
  Grade = grade,
  Count = count) |>
  st_drop_geometry()

# Convert dataframe to a table
grade_perc |>
  gt() |>
  tab_header(title = md("Percentage (%) of L.A. neighborhoods by HOLC grade"))
|>
  tab_options(table.width = pct(50), heading.title.font.size = 10)
```

Percentage (%) of L.A. neighborhoods by HOLC grade

Grade	Count	% of Census Blocks
A	449	5.0
B	1239	13.8
C	3058	34.0
D	1346	15.0
Ungraded	2896	32.2

Create two visualizations

Create at least two visualizations within HOLC grades:

- % low income
- percentile for Particulate Matter 2.5
- percentile for low life expectancy

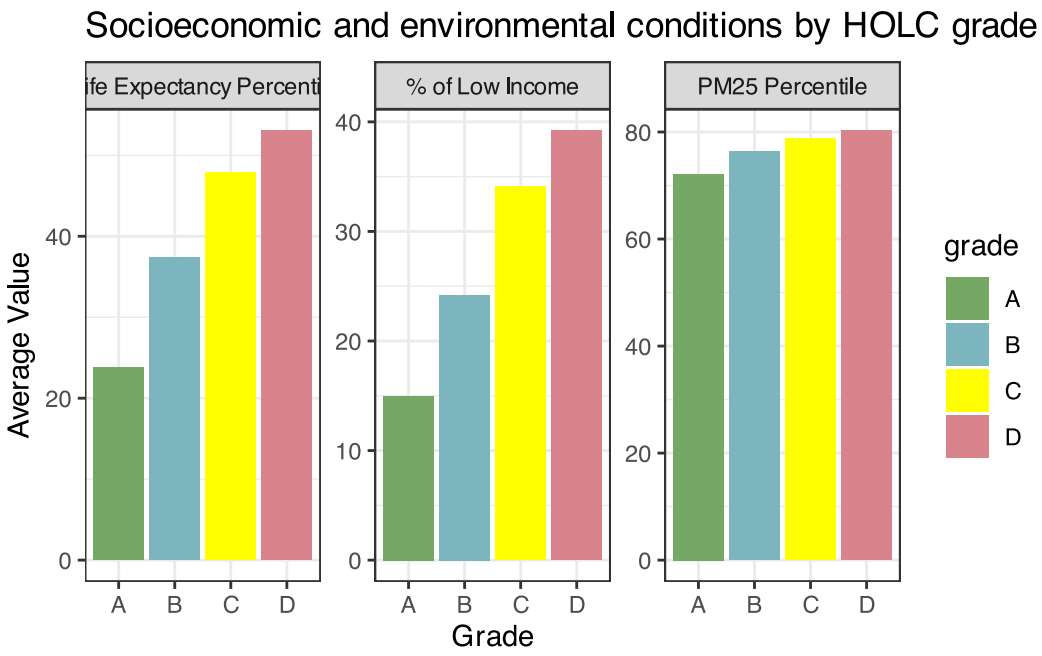
```
# Join LA and redline spatial data to create a summary
holc_join <- st_join(los_angeles, holc_transform,
join = st_intersects)
```

```
# Grouped per environemental conditions by grade
grouped_sum <- holc_join |>
  group_by(grade) |>
```

```
  summarise(
    mean_low_inc = mean(lowincpct, na.rm = TRUE) * 100,
    mean_pm25 = mean(p_pm25, na.rm = TRUE),
    mean_life_exp = mean(p_lifeexppct, na.rm = TRUE)
    ) |>
    group_by(grade) |>
    drop_na(grade) |>
    st_drop_geometry()
```

```
# Pivot longer to one column for all environmental conditions
 combined_groups <- grouped_sum |>
  pivot_longer(
    cols = c(mean_low_inc, mean_pm25, mean_life_exp),
    names_to = "condition_within_HOLC",
    values_to = "mean_percentage")
```

```
# Facet wrap by HOLC grades for each current variable conditions
ggplot(combined_groups, aes(x = grade, y = mean_percentage, fill = grade)) +
  geom_col() +
      facet_wrap(~condition_within_HOLC,   scales   =   "free_y",   labeller   =
as_labeller(c("mean_low_inc" = "% of Low Income",
  "mean_pm25" = "PM25 Percentile",
  "mean_life_exp" = " Life Expectancy Percentile"))) +
  labs(title= "Socioeconomic and environmental conditions by HOLC grade") +
  labs(x = "Grade", y = "Average Value") +
  scale_fill_manual(values = holc_color)
```



Interpretation

Discuss potential relationships between historical redlining grades and current environmental/ socioeconomic conditions:

While looking at the graphs, the group graded "D" had a higher average for all three conditions: percentage of low income, life expectancy percentile, and pm25 percentile. For the life expectancy, it shows roughly 60th percentile for those in group D meaning they have a lower life expectancy than 60% of those in the other neighboring area. The percentile lower with each grade and A has the smallest percentile. The low income is found to be at an average of 40% in group D. While the pm25 percentile was found roughly the same, since it could just mean environmental hazards could be refelcted in all neighborhoods despite the grading as those are independent factors such as the air in the area. The relationship could be that those who live these areas considered undesireable are the facing the environmental and socioeconomic factors that can be traced back to redlining.

Part 2: Legacy of redlining in pedestrian and bike accidents

```
# Read in Crash data
 crashes <- read_csv(here::here("data",
"SWITRS_LA", "Crashes.csv"))
```

```
# Filter through columns for NA
crash_bk_ped <- crashes |>
  filter(PEDESTRIAN_ACCIDENT == "Y" | BICYCLE_ACCIDENT == "Y") |>
```

```
  filter(POINT_X != 'NA')

# Count columns with NA that returns true
sum(crash_bk_ped$POINT_X == 'NA')
sum(crash_bk_ped$LATITUDE == 'NA')

# Transform data to a sf
crash_transform <- st_as_sf(crash_bk_ped, coords = c('POINT_X', 'POINT_Y'), crs
= 4326)
```

```
# Check crs for both sf
st_crs(crash_transform)
st_crs(holc_join)
```

```
# Check the matching crs
st_crs(crash_transform) == st_crs(holc_join) # returned false
```

```
# Transform to match crs
crash_object <- st_transform(crash_transform, st_crs(holc_join))
```

```
# Subset points
holc_crashes <- crash_object[holc_join, ]

# Left join
crash_left <- st_join(holc_crashes, holc_join)
```

```
#Create a table with grade and number of points filtering out NA's
crash_table <- crash_left |>
  filter(!is.na(grade)) |>
  group_by(grade) |>
  summarise(n_points = n()) |>
  st_drop_geometry()
```

```
# Mutate values to percentages
crash_perc <- crash_table |>
mutate(percentage = 100 * (n_points/sum(crash_table$n_points)))
# group by grade column as categorical
crash_perc$grade <- as.factor(crash_perc$grade)
# Convert to decimal for percentage
crash_perc$percentage <- as.double(crash_perc$percentage)
```

```r
# Plot the bike and pedestrian accident by grade and percentage
ggplot(crash_perc, aes(x = grade, y = percentage, fill = grade)) +
geom_col() +
scale_fill_manual(values = holc_color) +
labs(x = "Grade", y = "Percentage (%)") +
labs(title= "Bike and pedestrian accidents and fatalities by HOLC grade in
Los Angeles")
```

